Hybrid LLM-Guided Diagnosis Retrieval for Long Clinical Narratives

將臨床敘述, 轉換成類選擇題的診斷建議清單；可訓練的檢索模型縮小範圍, 再由固定的 LLM 判斷同義與排序, 不需 fine-tune LLM

作者: 吳定栩、陳冠宏，指導教授: 黃騰毅　國立臺灣科技大學 電機工程系

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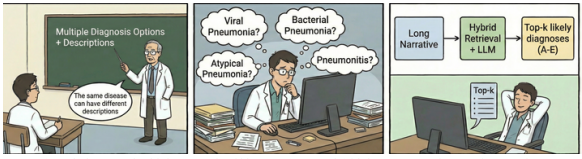
Clinical context and motivation

During training, medical students mostly see standardized, multiple-choice style case examples with clear labels and a small set of options. However, In real practice, clinicians face long & free-text case narratives where the same disease can be described in many different ways. This makes it hard for pure algorithms diagnosis to handle long text directly, so our goal is to turn these "fill in the blank" narratives into a considered set of options, giving clinicians clues on candidates diognosis they may missed.



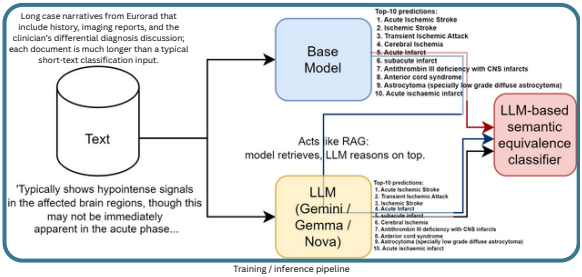
Training: multiple options, clear labels　Real world: long narratives, no explicit labels　Goal: support decision-making

Convert free-text diagnoses into a set of options for clinicians to review, rather than replacing their judgment.

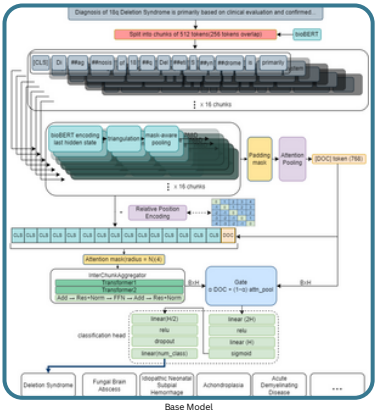
Research Objectives

- Build a trainable retrieval model for long case narratives that directly outputs a Top-k list of diagnosis candidates.
- Use a frozen large language model as a judge that only decides whether each candidate diagnosis is semantically equivalent to the gold disease, without fine-tuning the LLM, with require lots of resources.
- Combine the base retrieval model with the LLM judge into a hybrid pipeline that improves Top-k and MRR over single model, and is modular enough to deploy on in-hospital servers.

Methods and model architecture

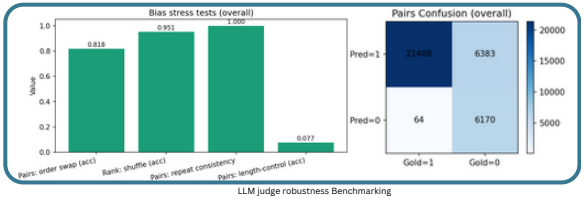


Training / inference pipeline



Base Model

Our base model first splits each clinical case into overlapping 512-token chunks, so that information near chunk boundaries is still captured by multiple chunks. Each chunk is encoded by a BioBERT encoder and converted into a chunk-level vector via token-attentive pooling. A sparse inter-chunk Transformer with relative position bias and a gating mechanism then aggregates all chunk vectors into a single case-level representation. Finally, a lightweight MLP classifier predicts diagnosis scores for all labels, and a simple class-similarity re-ranking step slightly adjusts these scores so that clinically confusable diagnoses receive more consistent rankings.

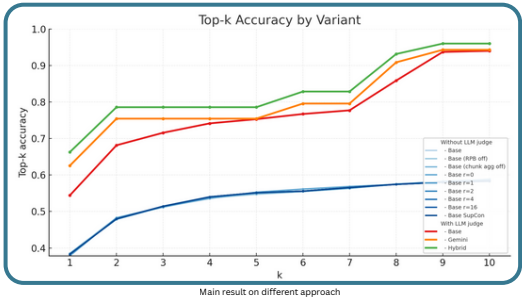


LLM judge robustness Benchmarking

This figure shows how reliable our MONDO-based LLM judge is. On the left, the bias stress tests show the judge stays accurate when we swap order, shuffle, or repeat pairs, but performance collapses in the length-control setting, revealing a strong bias toward string-length cues. On the right, the confusion matrix shows that most true synonym pairs are predicted as matches and most non-synonyms as non-matches, with few missed positives, so overall the judge is still usable as a semantic-equivalence signal.(Benchmarking on AWS NOVA pro)

Results

LLM-only can fix many errors caused by synonyms, and adding the Hybrid on top of the base retriever further improves Top-k and MRR.
The remaining ceiling mainly comes from how complete our synonym inventory is.



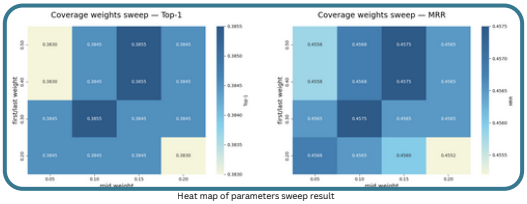
Main result on different approach

- Sweep over Base variants

Across all Base-only variants (turning RPB on/off, removing chunk aggregation, and setting radius r = 0/1/2/4/16 or SupCon+re-rank), performance is almost unchanged: Top-1 stays ~0.41, Top-10 around 0.63–0.64, and MRR ~0.49. The curves in the figure nearly overlap, indicating that these architecture and hyperparameter tweaks alone bring less than 1% gain and do not meaningfully improve retrieval quality.

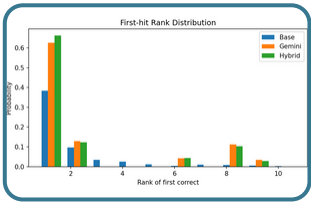
- Base vs. Gemini vs. Hybrid

In contrast, the three main systems show clear separation: Base reaches Top-1 ~ 0.56 and Top-10 ~ 0.94, Gemini improves to about Top-1 ~ 0.65 and Top-10 ~ 0.95, and the Hybrid model (Base + Gemini re-ranking) further pushes Top-1 to ~ 0.69 and Top-10 to ~ 0.97. The plotted curves highlight that "Base to narrow candidates, then Gemini to re-rank" gives the most accurate rankings, especially for small k.



Heat map of parameters sweep result

- Setup: The heatmaps sweep coverage weights: x-axis is the mid-chunk weight, y-axis is the shared weight for first/last chunks.
- Results: While overall metrics stay stable, the best scores appear when emphasizing the narrative boundaries.
- Key Finding: Optimal performance occurs with first/last weights ~ 0.15 and mid weight (~ 0.15), confirming that diagnostic clues are concentrated at the beginning and end.



Base often makes "near-miss" errors, with a correct synonym sitting just below the top choice (16.75% at rank 2, 8.13% at rank 3). The first-hit rank curves show rank-1 hits improving from 38.29% (Base) to 62.57% (Gemini) and 66.29% (Hybrid), meaning Gemini and especially Hybrid mainly fix these near misses and move the right diagnosis into the top position.

Conclusion

- Bridge the Gap: Successfully converts unstructured, long clinical narratives into actionable Top-k diagnosis options.
- Hybrid Advantage: Combines the structural efficiency of a retriever with the semantic reasoning of frozen LLMs, outperforming single-stage models.
- Privacy-First: The modular design supports on-premise deployment using local models, ensuring no PHI leaves the hospital.

Future work

- Weak Supervision: Expand the limited training set using LLM-augmented synonym expansion to improve retriever calibration.
- Active Learning: Integrate a clinician feedback loop to dynamically update the synonym inventory and refine the model over time.
- Generative Ranking: Explore full generative re-ranking (e.g. AWS Nova / Gemini) to compare against the current T/F judgment approach.

This work has been selected for poster presentation at the 4th IPAS International Symposium (Organized by NTUST & Yonsei University).